Network Layer (4)

COMP90007 Internet Technologies

Lecturer: Adel N. Toosi

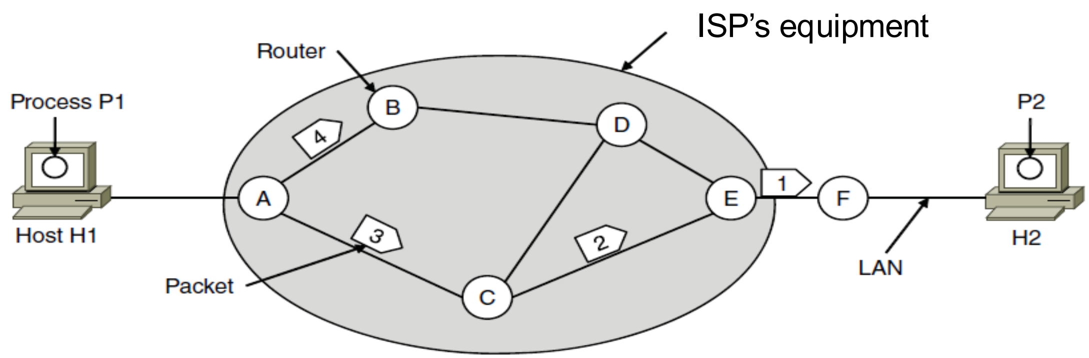
Semester 1, 2026

Outline

- Network layer on the Internet
- Types of services
- Internetworking
  - Tunneling
  - Fragmentation
  - Path MTU discovery
- Internet Protocol
  - Addressing
  - Subnetting
- Routing algorithms

Routing

- Routing is the process of discovering network paths
    - Consider the network as a graph of nodes and links
    - Optimisation metrics: hops, delay, etc.
    - Update routes for changes in topology (e.g., router failures)



A's table (initially)		A's table (later)		C's Table		E's Table	
A	–	A	–	A	A	A	C
B	B	B	B	B	A	B	D
C	C	C	C	C	–	C	C
D	B	D	B	D	E	D	D
E	C	E	B	E	E	E	–
F	C	F	B	F	E	F	F

Dest. Line

Routing Algorithms (1)

- The routing algorithm determines the **appropriate output line for forwarding** an incoming packet

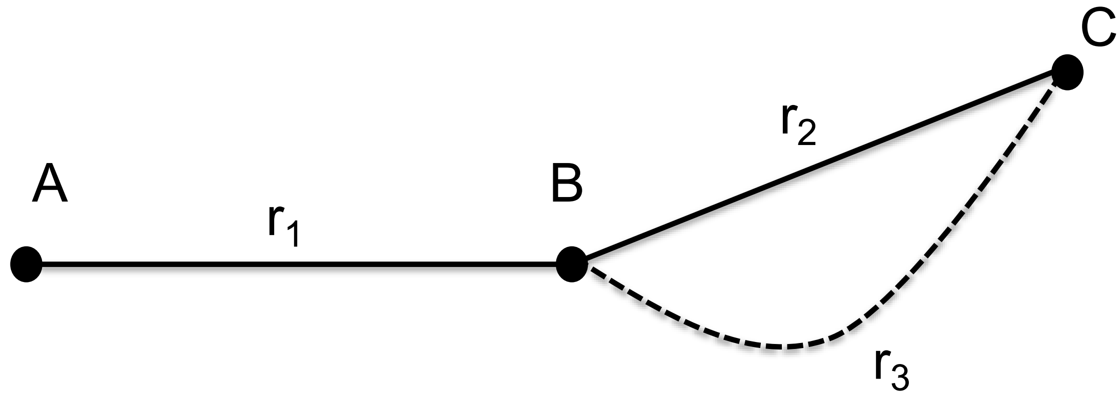
- **Non-Adaptive Algorithms**
  - Static routing, static decision-making process
- **Adaptive Algorithms**
  - Dynamic decision-making process
  - Changes in network topology, traffic, etc.

Routing Algorithms (2)

- Non-adaptive
  - Shortest path routing
  - Flooding
- Adaptive
  - Distance vector routing
  - Link state routing
- Hierarchical routing
- Broadcasting routing

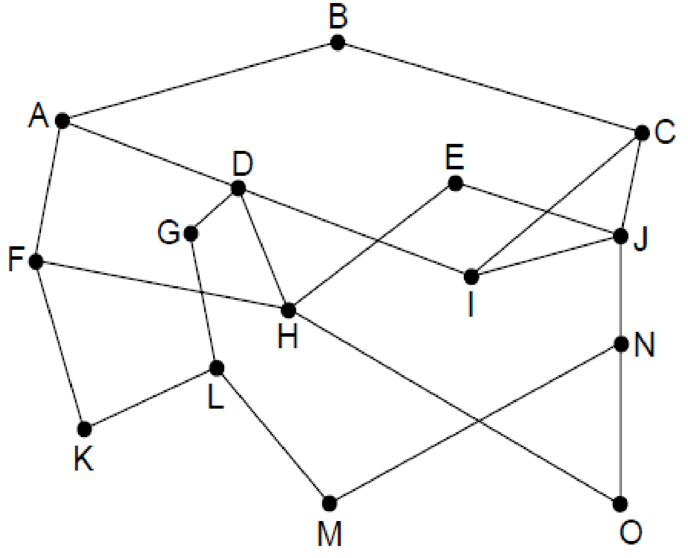
Optimality Principle

- If router B is on the optimal path from router A to router C, then the optimal path from B to C also falls along the same route.

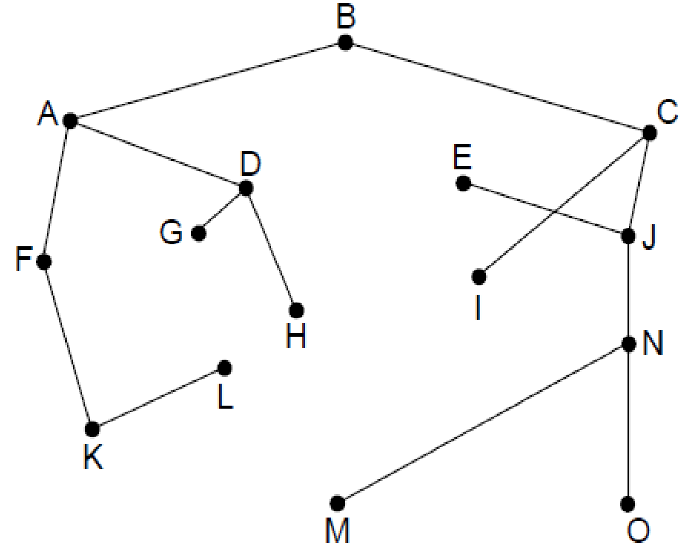


Sink Tree

- **Sink Tree**: the set of optimal routes from all sources to a given destination forms a tree rooted at the destination
- **Goal** of a routing algorithm: to discover and utilise the sink trees for each router to select optimal paths



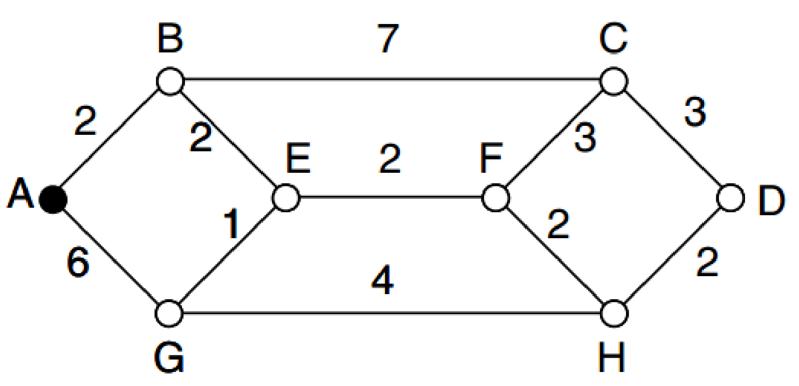
Network



Sink tree for router B

Shortest Path Routing

- Non-adaptive algorithm
- The algorithm finds the shortest path between two routers on the graph
- Metrics: number of hops, distance, delay etc.

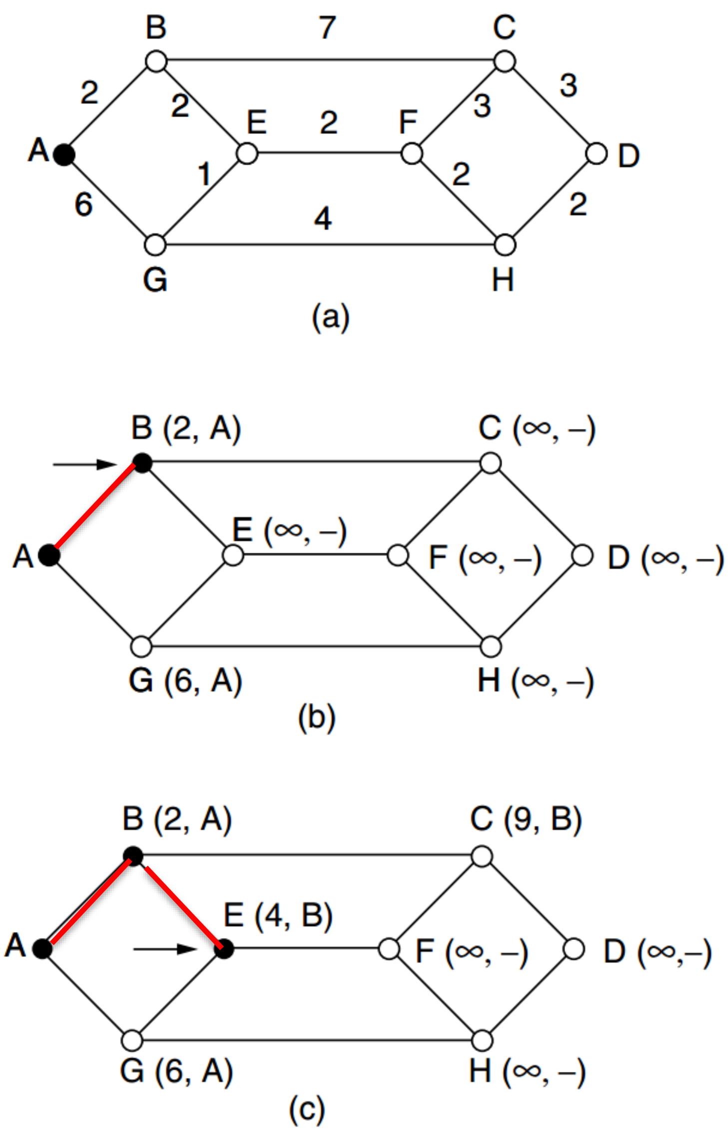


- Solution: compute a sink tree on the graph
  - Each link is assigned a non-negative weight/distance
  - The shortest path has the lowest total weight
  - Using weight of 1 on each link gives the path with fewest hops

Shortest Path: Dijkstra's Algorithm

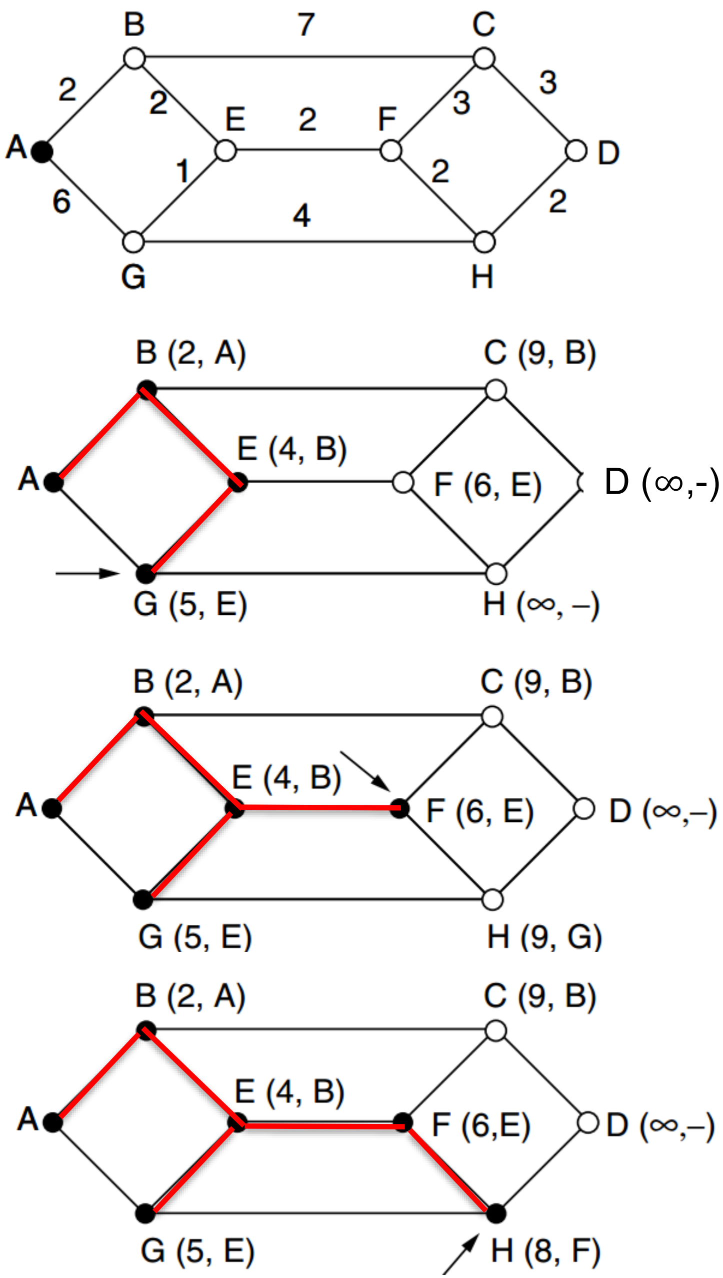
1) Create **a set *P, tracking* the nodes added in the tree**. Initialise it as empty.
2) For each node, assign a **distance value *d* from the node to the sink node**. Initialise *d* for all nodes as infinity.
3) Start from the sink node, assign distance as 0*.*
4) **Repeat** when *P* doesn't include all nodes:
   i. For all the nodes not in *P,* compare distance *d.*
   ii. Pick a node *v* with min distance and add it to *P.*
   iii. Update *d* for all the adjacent nodes of *v* (newly added node).

Example: Dijkstra's Algorithm



Distance to A **Set P**

n	A	B	C	D	E	F	G	H	Set P
1	0	∞	∞	∞	∞	∞	∞	∞	{A}
2	--	**2**	∞	∞	∞	∞	6	∞	{A, B}
3	--	--	9	∞	**4**	∞	6	∞	{A, B, E}



. . .

Distance to A — Set P

n	A	B	C	D	E	F	G	H	Set P
1	0	∞	∞	∞	∞	∞	∞	∞	{A}
2	--	2	∞	∞	∞	∞	6	∞	{A, B}
3	--	--	9	∞	4	∞	6	∞	{A, B, E}
4	--	--	9	∞	--	6	**5**	∞	{A, B, E, G}
5	--	--	9	∞	--	**6**	--	9	{A, B, E, G, F}
6	--	--	9	∞	--	--	--	**8**	{A, B, E, G, F, H}
7	--	--	**9**	10	--	--	--	--	{A, B, E, G, F, H, C}
8	--	--	--	**10**	--	--	--	--	{A, B, E, G, F, H, C, D}

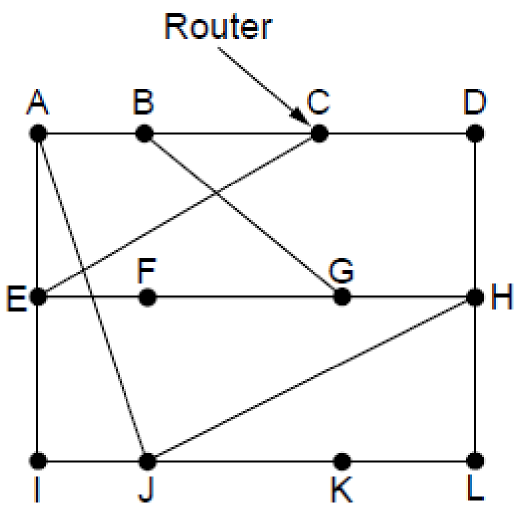
Flooding

- Non-adaptive algorithm
- Every incoming packet is sent out on **every outgoing line except the one on which it arrived**
- Inefficient: generates many duplicate packets
- Selective flooding is an improved variation
  - Routers send packets only on the links which are approximately in the right direction

Distance Vector Routing (1)

- Dynamic algorithm
  - Each router maintains a table which includes the lowest known distance to each destination and which line to use to get there
  - Tables are updated by exchanging information with neighbouring routers
  - Global information shared locally
- Algorithm: *each* router
  1) knows the distances to its neighbours
  2) **advertises** a vector of its lowest known distances to **all neighbours**
  3) **update** its own distance vector based on the received information
  4) repeats periodically to reflect network changes

Distance Vector Routing (2)



Network

JA = 8, JI = 10, JH = 12, JK = 6

To	A	I	H	K	New estimated delay from J	Line
A	0	24	20	21	8	A
B	12	36	31	28	20	A
C	25	18	19	36	28	I
D	40	27	8	24	20	H
E	14	7	30	22	17	I
F	23	20	19	40	30	I
G	18	31	6	31	18	H
H	17	20	0	19	12	H
I	21	0	14	22	10	I
J	9	11	7	10	0	–
K	24	22	22	0	6	K
L	29	33	9	9	15	K

Vectors received from neighbors A, I, H and K

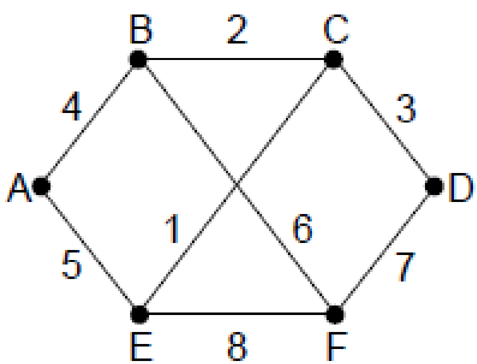
New vector for J

Link State Routing

- Dynamic algorithm
  - An alternative to distance vector: **too long to converge** after the network topology changed
  - Widely used on the Internet, e.g. Open Shortest Path First (OSPF)
  - Local information shared globally, using flooding
  - Requires more computation than the distance vector approach
- Algorithm: *each* router
  1) discovers neighbours and learn network addresses
  2) measures distances to each neighbour
  3) builds a **link state packet**
  4) sends this packet to **all the other routers**
  5) **computes the shortest path** to every other router, e.g. using Dijkstra's algorithm

Building Link State Packets

- The Link State Packet (LSP) for a node lists its neighbours and the distances to reach them



Network

Link State Packets

A		B		C		D		E		F	
Seq.		Seq.		Seq.		Seq.		Seq.		Seq.	
Age		Age		Age		Age		Age		Age	
B	4	A	4	B	2	C	3	A	5	B	6
E	5	C	2	D	3	F	7	C	1	D	7
		F	6	E	1			F	8	E	8

LSP for all nodes

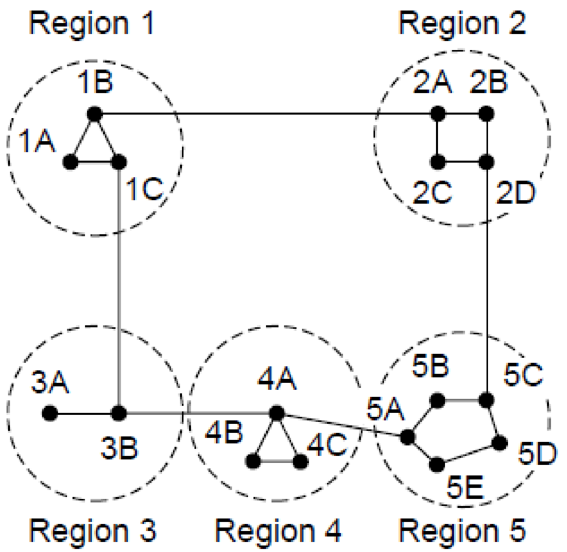
- When to build new LSP?
  - Periodically at regular intervals
  - Build them when some significant event occurs

Hierarchical Routing (1)

- As networks grow, routing tables increase in size, leading to higher CPU and memory demands
- Dividing all routers into regions
  - Each router knows everything about other routers in its region but nothing about the routers in other regions
  - Routers connected to two regions act as exchange points for routing decisions

Hierarchical Routing (2)

- Hierarchical routing reduces the computation but may result in slightly longer paths than flat routing



Full table for 1A

Dest.	Line	Hops
1A	–	–
1B	1B	1
1C	1C	1
2A	1B	2
2B	1B	3
2C	1B	3
2D	1B	4
3A	1C	3
3B	1C	2
4A	1C	3
4B	1C	4
4C	1C	4
5A	1C	4
5B	1C	5
5C	1B	5
5D	1C	6
5E	1C	5

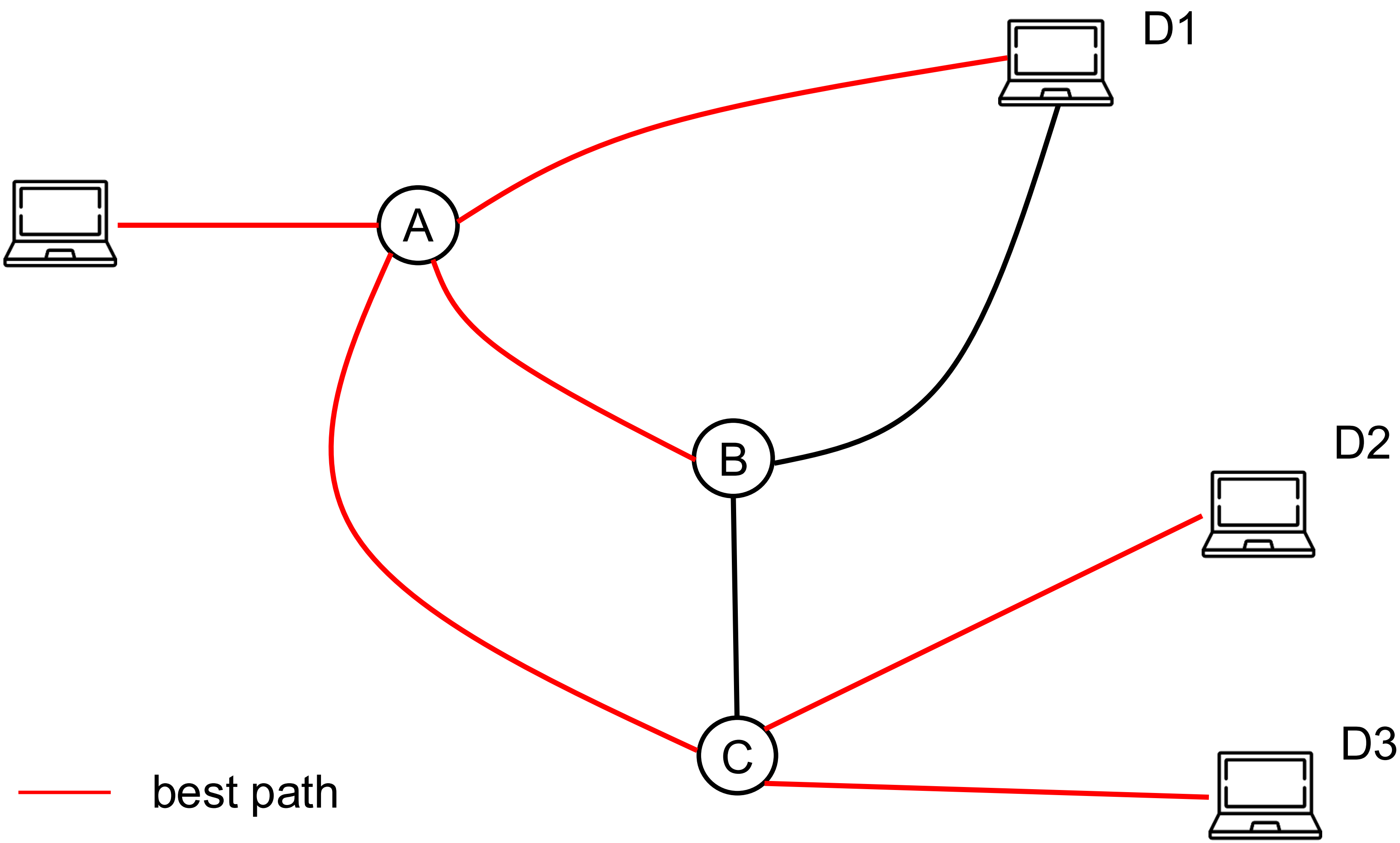
Hierarchical table for 1A

Dest.	Line	Hops
1A	–	–
1B	1B	1
1C	1C	1
2	1B	2
3	1C	2
4	1C	3
5	1C	4

Broadcast Routing

- Broadcast routing allows hosts to send messages to all other hosts
  - **Single distinct packet** to each destination
    - Inefficient, and the source needs all destination addresses
  - **Multi-destination routing**: a router copies the packet for each outgoing line
    - Use bandwidth more efficiently, but the source needs to know all destination addresses
  - **Flooding**
  - **Reverse path forwarding**

Broadcast Routing Examples



Reverse Path Forwarding

- The router checks if the packet is arrived on the line normally used for sending packets to the source of the broadcast
  - **Yes**: the router copies the packet and forwards them onto all other lines, due to a **high probability** that the route used to transmit this packet is the **best,** and this packet is the first copy.
  - **No:** the packet is **discarded** as it is likely a **duplicate**.

Summary

- Network layer on the Internet
- Types of services
- Internetworking
  - Tunneling
  - Fragmentation
  - Path MTU discovery
- Internet Protocol
  - Addressing
  - Subnetting
- Routing algorithms